

BIOL 413 Principals of Biology, Fall 2018

Professor: Dr. Sue Cooke, email: suzanne.cooke@unh.edu
Office/Phone: P475, 603-641-4328
Office Hours: Mondays, Wednesdays 12-1pm or by appointment
Class meets: Monday 1-3:50pm

Textbook:

Biology 2e from OpenStax, Print ISBN 1947172514, Digital ISBN 1947172522,
www.openstax.org/details/books/biology-2e

Your textbook for this class is available for free online! If you prefer, you can also get a print version at a very low cost. Your book is available in web view and PDF for free. You can also choose to purchase on iBooks or get a print version from OpenStax on Amazon.com. You can use whichever formats you want. Web view is recommended -- the responsive design works seamlessly on any device. If you buy on Amazon, make sure you use the link on your book page on openstax.org so you get the official OpenStax print version. (Simple printouts sold by third parties on Amazon are not verifiable and not as high-quality.)

Other Materials:

- Hofmann, A, H. 2016. Writing in the Biological Sciences, 2nd edition. Oxford University Press. ISBN 9780190245603.
- A lab coat is required and may be purchased through the UNH-M bookstore.
- You do not need to purchase a lab manual.

Classroom Policies

Evaluation:

Your overall course grade will be determined by calculating the percentage of points earned out of the possible 525.

3 lecture exams (100pts each)	300 points
Best 10/13 quizzes (10pts each)	100 points
Laboratory	<u>125 points</u>
	525 points

Your overall course grade will be determined by calculating the percentage of points earned out of the possible 525. **The grade scale is:** 94-100%=A; 90-93%=A-; 87-89%=B+; 84-86%=B; 80-83%=B-; 76-79%=C+; 73-75%=C; 70-72%=C-; 66-69%=D+; 63-65%=D; 60-62%=D; Below 59.99%=F.

Exams and Quizzes:

During the course you will take 3 exams. There will be no makeup exams without a documented excuse. If possible, speak to me in advance about conflicts with the course in general and specifically with scheduled exams. All exams are non-cumulative.

Weekly quizzes will be given on the material from the previous class. Quizzes will be taken online through our MyCourses site and must be completed by their due date to receive credit. The lowest 3 quiz grades will be dropped. This includes zeros from quizzes that were not completed by their due date. No exceptions will be given to this policy.

Extra credit policy:

There will be extra credit points available on tests. I may occasionally offer extra credit assignments and these will be optional. I will not give extra credit assignments to individual students who are unhappy with their grades because that is not fair to the rest of the class!

Attendance:

Attendance is mandatory. I will take attendance at the start of each class. If you arrive late, you are responsible for checking in with me before leaving class. This is your responsibility. I will not accept emails after class as proof of attendance! You should make every effort to be on time for class because walking in late is disturbing to other students.

Classroom behavior:

You are expected to conduct yourself in a professional manner at all times while in lecture or lab. Aside of participating and arriving on time, this also includes being respectful of the instructor and your fellow students. If you are being disruptive in any way, you will be asked to leave the classroom.

Please turn off your cell phones in class! If you MUST have your cell phone on because you are expecting an emergency call or need to be available for personal matters, please keep your cell phone on vibrate and leave the room to answer it. If possible, inform me in advance if you are expecting a call and sit on the end of a row so you minimize disturbances to your fellow students! DO NOT TEXT OR ACCESS FACEBOOK OR OTHER SOCIAL MEDIA DURING CLASS.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4383, Jenessa.Zurek@unh.edu, or in person in the Student Services Suite.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<https://manchester.unh.edu/current-students/academic-counseling#student-rights>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Academic Support/Tutoring:

The Center for Academic Enrichment is a resource for all students enrolled in UNH Manchester courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for you class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor.

Course Schedule – Fall 2018

The following schedule is tentative and subject to change based on how much time we need to cover the material. Effort will be made to announce any changes to this schedule as far in advance as possible.

<u>Date</u>	<u>Topic</u>	<u>Chapter</u>
Aug. 27 th	Introduction, The Study of Life	1
Sept. 3 rd	Labor Day...No face to face classes Virtual lecture: Chemical Foundation of Life	2
Sept. 10 th	Biological Macromolecules	3
Sept. 17 th	Cell Structure	4
Sept. 24 th	Structure & Function of Plasma Membranes	5
Oct. 1st	Exam 1 (Chapters 1-5)	
Oct. 8 th	UNHM Fall Break...no face to face classes Virtual lecture: Cell Reproduction	10,11
Oct. 15 th	Heredity	12,13
Oct. 22 nd	DNA Structure and Function	14
Oct. 29 th	Gene Expression	15, 16
Nov. 5th	Exam 2 (Chapters 10-16)	
Nov. 12 th	Veteran's Day...no face to face classes Virtual lecture: Biotechnology and Genomics	17
Nov. 19 th	Metabolism	6
Nov. 26 th	Cellular Respiration	7
Dec. 3 rd	Photosynthesis	8
Dec. 10th	Exam 3 (Chapters 17, 6-8)	

BIOL 413 Laboratory

Fall 2018

Contact Information:

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Office:

Rm 550B

Feel free to contact me at any time. Email works best, I will attempt to respond to you as soon as possible.

Lab Schedule:

Module 1: Scientific Method

Aug 29, 30, 31	Safety, Introduction to Brine Shrimp and Scientific Method
Sep 5, 6, 7	Experimental Design
Sep 12, 13, 14	No Labs
Sep 19, 20, 21	Run Experiment, Data Collection, Statistical Analysis
Sep 26, 27, 28	Communicating in Science; Anatomy of a Journal Article, Conventions, Presentations **Quiz 1 <i>brine shrimp</i>
Oct 3, 4, 5	Student Presentations

Module 2: Cells

Oct 10, 11, 12	Mitosis, Meiosis and Cell Division
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Module 3: Molecular Genetics

Oct 17, 18, 19	Crime Scene Evidence Collecting & Blood Typing **Quiz 2 Mitosis/Meiosis
Oct 24, 25, 26	Crime Scene ELISA Test **Quiz 3 Blood Typing
Oct 31, Nov 1, 2	Crime Scene Restriction Fragment Analysis ** Quiz 4 ELISA
Nov 7, 8, 9	Crime Scene Final Analysis & Communicating in Science – Forensic report ** Quiz 5 RFLP
Nov 14, 15, 16	Student Presentations
Nov 21, 22, 23	Thanksgiving No Labs

Module 4: Cellular Respiration and Metabolism

Nov 28, 29, 30	Metabolism
Dec 5, 6, 7	Lab Exam

Grading:

Lab grade will be a total of 125pts made up of the following:

Presentation/Lab Report from Module 1	20pts
Presentation/Lab Report from module 3	20pts
Lab Report from Module 2	10pts
Lab Report from module 4	10pts
Lab Quizzes	25pts
Lab Exam	20pts
Participation/Attendance	20pts
Total	125pts

Participation/Attendance:

Attendance in lab is very important as each lab of a module builds upon another. If you miss a lab you will be responsible for gathering the information needed to come to the next lab ready to perform the lab of that week. Also we will be working as groups during this lab and it is not fair to the others in your group to miss lab as that will leave the other members of the group more work to do. In light of these facts **5 pts will be deducted from your lab grade for each missed lab**. Any more than 2 missed labs will result in possible failure of the lab. Also all within a group must contribute to the assignments in this lab or points may be deducted from those who don't equally participate.

Canvas (My Courses):

All of the materials for this lab are found in your lab section of Canvas. While there may be times where you will be given hard copies of materials you will need for lab there will also be times where you will need to print out and bring the materials to lab. It is also a good way for you to review any videos or interactive programs that we may use in lab.

Electronic Devices:

You are welcome to use electronic devices (cell phones, tablets, laptops) in lab for "lab related" purposes. If you however need, or any other non-lab related use of these devices (i.e. to take or place a phone call) please be respectful of your lab mates and go outside the lab for such activities.

Lab Notebook:

You will need a notebook for lab to record your experimental progress and data in.

Lab Manual:

There is no lab manual that needs to be purchased for this lab.